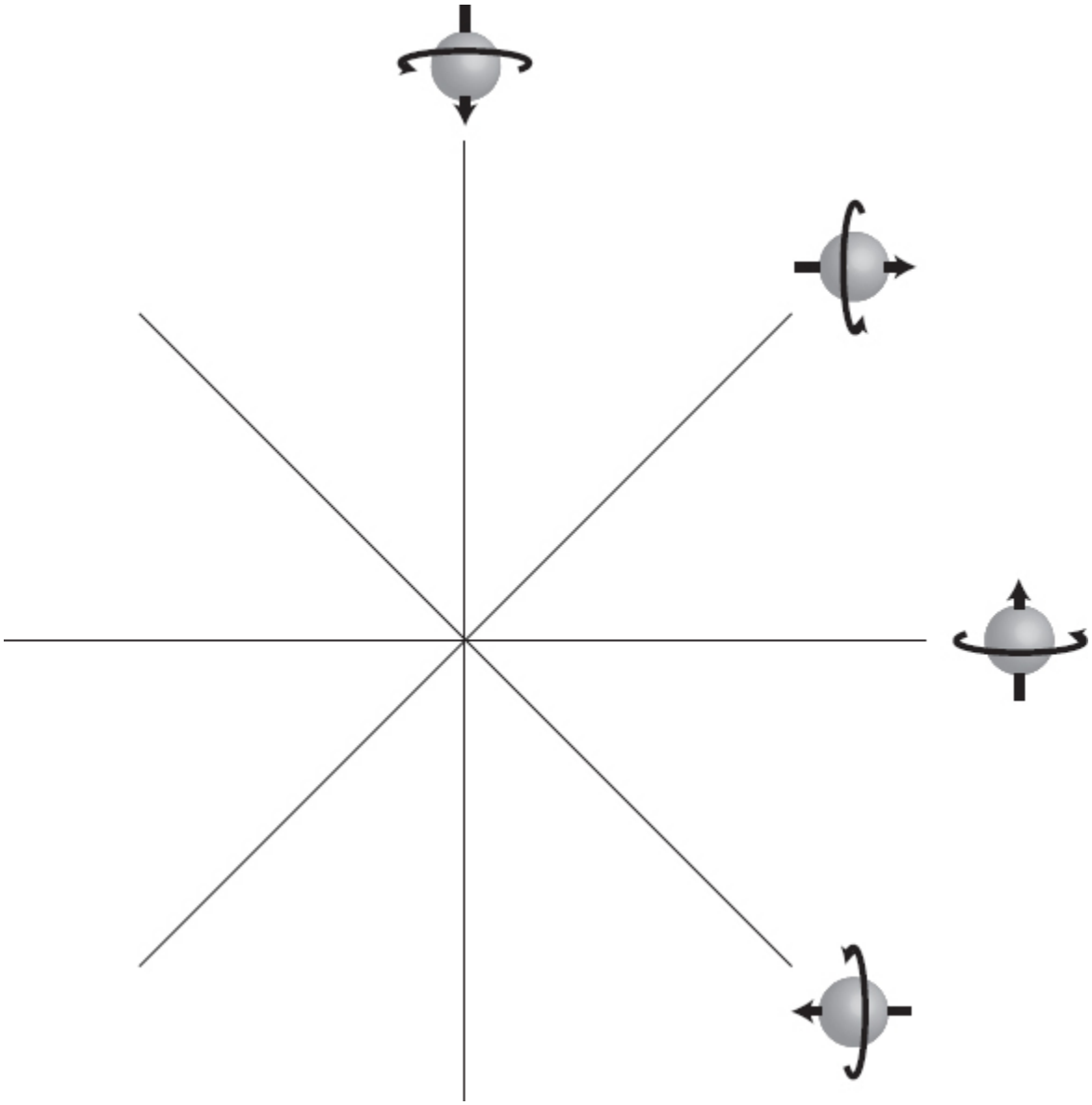


Now we can see where the uncertainty principle comes from. For a single spin, the uncertainty principle says that the state can't have a definite value for the spin along the original axes (up/down) and the rotated axes (right/left) at the same time. This is clear from the picture: if the state is purely spin-up, it's automatically some combination of spin-left and spin-right, and vice versa.

Just as there are no quantum states that are simultaneously localized in position and momentum, there are no states that are simultaneously localized in both vertical spin and horizontal spin. The uncertainty principle reflects the relationship between what really exists (quantum states) and what we can measure (one observable at a time).